ASSIGNMENT- 02

Bachelor of Computer Applications (Hons. AI and ML)
Semester- III

Subject: Database Management System

Topic: ER Diagram Design and Relational Schema

Application: Online Food Delivery System

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ER Diagram and Relational Schema

Application: Online Food Delivery System (modelled on platforms such as Swiggy or Zomato).

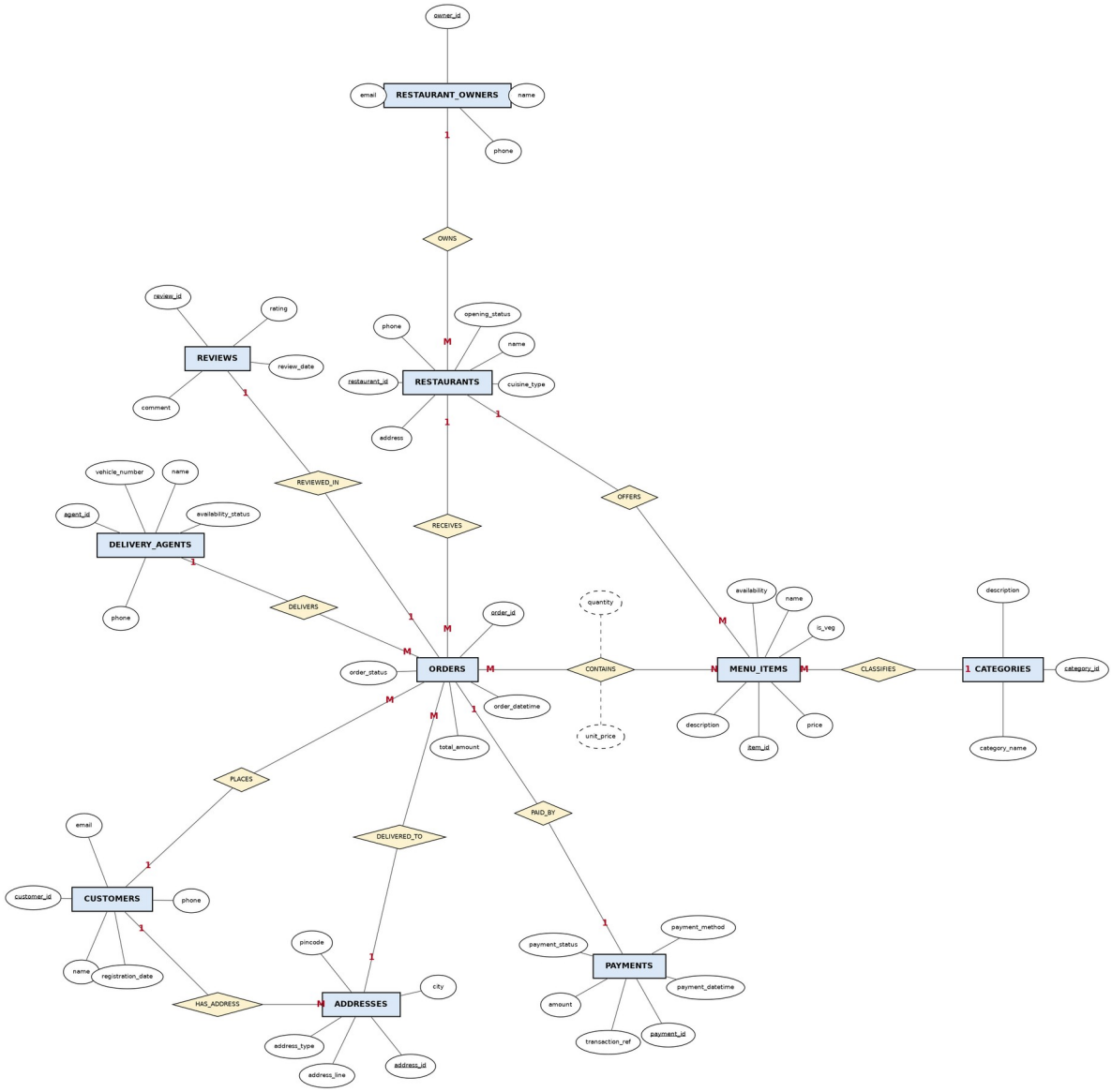
Purpose: Registered customers browse restaurants, view menus, place orders, pay online and receive the food at a chosen address through a delivery agent. Restaurant owners manage their menu items. The database tracks the whole order lifecycle (order, payment, delivery agent assignment, delivery) and the customer's rating and review after delivery, while keeping data consistent and avoiding repetition.

1. List of Entities and Attributes

Primary keys are marked (PK). Foreign keys are marked (FK).

No.	Entity	Attributes
1	**CUSTOMERS**	customer_id (PK), name, email, phone, registration_date
2	**ADDRESSES**	address_id (PK), address_line, city, pincode, address_type, customer_id (FK)
3	**RESTAURANT_OWNERS**	owner_id (PK), name, email, phone
4	**RESTAURANTS**	restaurant_id (PK), name, phone, address, cuisine_type, opening_status, owner_id (FK)
5	**CATEGORIES**	category_id (PK), category_name, description
6	**MENU_ITEMS**	item_id (PK), name, description, price, is_veg, availability, restaurant_id (FK), category_id (FK)
7	**DELIVERY_AGENTS**	agent_id (PK), name, phone, vehicle_number, availability_status
8	**ORDERS**	order_id (PK), order_datetime, order_status, total_amount, customer_id (FK), restaurant_id (FK), address_id (FK), agent_id (FK)
9	**ORDER_ITEMS**	order_id (PK, FK), item_id (PK, FK), quantity, unit_price
10	**PAYMENTS**	payment_id (PK), amount, payment_method, payment_status, transaction_ref, payment_datetime, order_id (FK)
11	**REVIEWS**	review_id (PK), rating, comment, review_date, order_id (FK)

2. ER Diagram



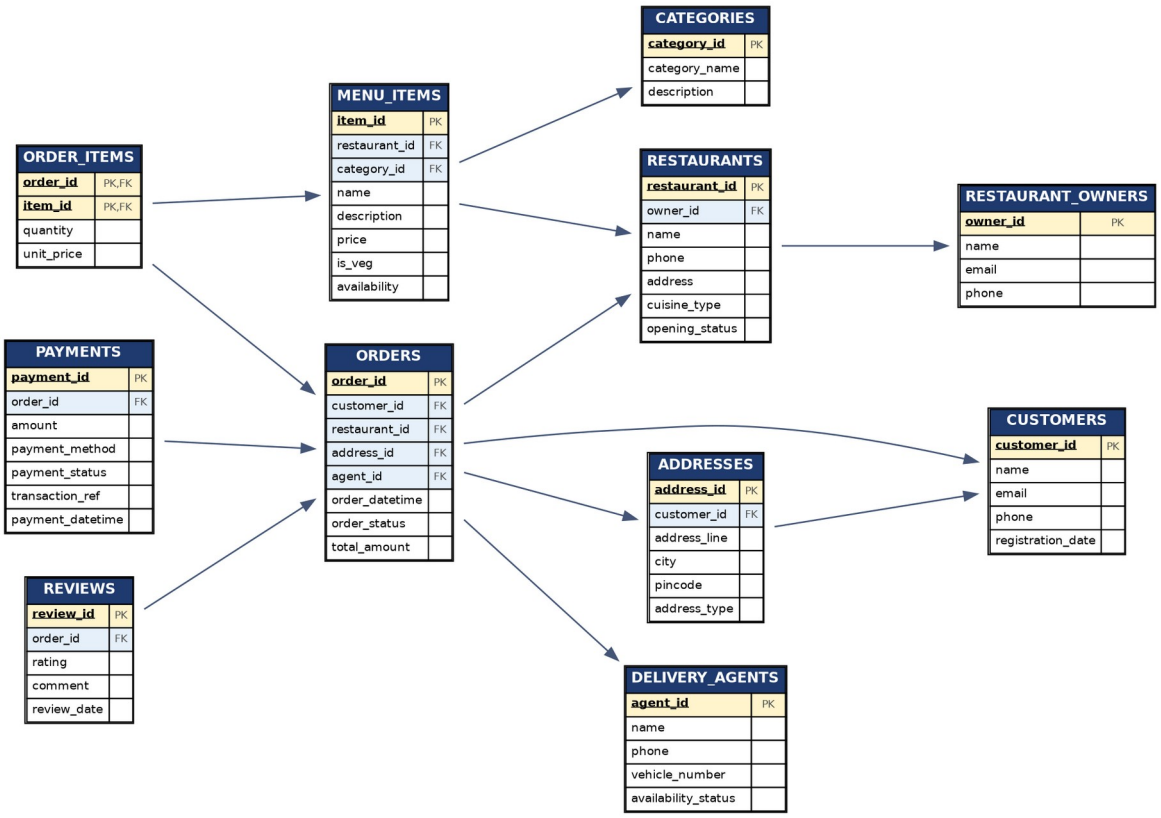
Notation:

- **rectangles** are entities,
- **ellipses** are attributes (underlined = primary key; dashed = attribute of a relationship),
- **diamonds** are relationships,
- **1, M and N** beside each relationship show its cardinality.

3. Relationships and Cardinalities

Relationship	Entity 1	Entity 2	Cardinality
PLACES	CUSTOMERS	ORDERS	One-to-Many (1:M)
HAS_ADDRESS	CUSTOMERS	ADDRESSES	One-to-Many (1:M)
DELIVERED_TO	ADDRESSES	ORDERS	One-to-Many (1:M)
OWNS	RESTAURANT_OWNERS	RESTAURANTS	One-to-Many (1:M)
RECEIVES	RESTAURANTS	ORDERS	One-to-Many (1:M)
OFFERS	RESTAURANTS	MENU_ITEMS	One-to-Many (1:M)
CLASSIFIES	CATEGORIES	MENU_ITEMS	One-to-Many (1:M)
DELIVERS	DELIVERY_AGENTS	ORDERS	One-to-Many (1:M); agent is assigned after order is accepted
PAID_BY	ORDERS	PAYMENTS	One-to-One (1:1); payment is recorded when the customer pays
REVIEWED_IN	ORDERS	REVIEWS	One-to-One (1:1); a review is optional
CONTAINS	ORDERS	MENU_ITEMS	Many-to-Many (M:N)

4. Relational Schema



Yellow cells are primary keys and blue cells are foreign keys. Arrows point from the table holding the foreign key to the table it references.

5. Relations (Table)

Table	Attributes
CUSTOMERS	customer_id, name, email, phone, registration_date
ADDRESSES	address_id, customer_id, address_line, city, pincode, address_type
RESTAURANT_OWNERS	owner_id, name, email, phone
RESTAURANTS	restaurant_id, owner_id, name, phone, address, cuisine_type, opening_status
CATEGORIES	category_id, category_name, description
MENU_ITEMS	item_id, restaurant_id, category_id, name, description, price, is_veg, availability
DELIVERY_AGENTS	agent_id, name, phone, vehicle_number, availability_status
ORDERS	order_id, customer_id, restaurant_id, address_id, agent_id, order_datetime, order_status, total_amount
ORDER_ITEMS	order_id, item_id, quantity, unit_price
PAYMENTS	payment_id, order_id, amount, payment_method, payment_status, transaction_ref, payment_datetime
REVIEWS	review_id, order_id, rating, comment, review_date

The first attribute(s) listed for each table form its primary key.

Conversion rules used

- Each entity became a table, and its key attribute became the primary key.
- For every 1:M relationship, the primary key of the “one” side was added as a foreign key in the “many” side table.
- The 1:1 relationships ORDERS–PAYMENTS and ORDERS–REVIEWS place order_id in PAYMENTS and REVIEWS as a unique foreign key.
- The M:N relationship between ORDERS and MENU_ITEMS became the associative table ORDER_ITEMS, whose composite primary key is made of the two foreign keys; the relationship attributes quantity and unit_price were moved into it.

6. Primary and Foreign Key Identification

Table	Primary Key	Foreign Key	References
CUSTOMERS	customer_id	–	–
ADDRESSES	address_id	customer_id	CUSTOMERS(customer_id)
RESTAURANT_OWNERS	owner_id	–	–
RESTAURANTS	restaurant_id	owner_id	RESTAURANT_OWNERS(owner_id)
CATEGORIES	category_id	–	–
MENU_ITEMS	item_id	restaurant_id category_id	RESTAURANTS(restaurant_id) CATEGORIES(category_id)
DELIVERY_AGENTS	agent_id	–	–
ORDERS	order_id	customer_id restaurant_id address_id agent_id (nullable)	CUSTOMERS(customer_id) RESTAURANTS(restaurant_id) ADDRESSES(address_id) DELIVERY_AGENTS(agent_id)
ORDER_ITEMS	order_id, item_id (composite)	order_id item_id	ORDERS(order_id) MENU_ITEMS(item_id)
PAYMENTS	payment_id	order_id (unique)	ORDERS(order_id)
REVIEWS	review_id	order_id (unique)	ORDERS(order_id)

7. Brief Relationship Explanation

- **CUSTOMERS – ORDERS (PLACES, 1:M):** One customer can place many orders over time, but every order is placed by exactly one customer.
- **CUSTOMERS – ADDRESSES (HAS_ADDRESS, 1:M):** A customer can save several delivery addresses (home, work, etc.); each saved address belongs to one customer.
- **ADDRESSES – ORDERS (DELIVERED_TO, 1:M):** An address can be used for many orders, while each order is delivered to the one address chosen at checkout.
- **RESTAURANT_OWNERS – RESTAURANTS (OWNS, 1:M):** An owner can run many restaurant outlets; each restaurant is managed by one owner, who maintains its menu.
- **RESTAURANTS – ORDERS (RECEIVES, 1:M):** A restaurant receives many orders, and each order is placed with a single restaurant.
- **RESTAURANTS – MENU_ITEMS (OFFERS, 1:M):** A restaurant offers many menu items; each menu item belongs to one restaurant.
- **CATEGORIES – MENU_ITEMS (CLASSIFIES, 1:M):** A category (for example Starters, Biryani, Desserts) groups many menu items, and each item falls under one category.
- **DELIVERY_AGENTS – ORDERS (DELIVERS, 1:M):** A delivery agent delivers many orders over time, but each order is delivered by one agent. The agent_id stays empty (NULL) until an agent is assigned.
- **ORDERS – PAYMENTS (PAID_BY, 1:1):** Each order is settled by one payment record, and each payment belongs to exactly one order.

- **ORDERS – REVIEWS (REVIEWED_IN, 1:1):** After delivery, the customer may submit one rating and review for the order. Since the customer and restaurant can be found through the order, they are not repeated in REVIEWS.
- **ORDERS – MENU_ITEMS (CONTAINS, M:N):** An order can contain several menu items, and a menu item can appear in many orders. This is resolved by the ORDER_ITEMS table, which also stores quantity and unit_price (the price at the time of ordering).